Primer: AIRL with Unobserved Heterogeneity (AIRL-Het)

Lee, Sudhir and Wang (2026)

Setup

Actions. $\mathcal{A} = \{0, 1, 2\}$ corresponding to pay (immediate access), wait (delayed free access), and exit.

State. $s_{ijt} = (C_{ijt},\ g_{ijt},\ P^0_{ijt},\ P^1_{ijt},\ w_j,\ \texttt{pay}_{ij(t-1)},\ X_{ijt})$ where C is a 32-dim content embedding, g is progress through the series, P^0, P^1 are price indicators, w_j is the series-level wait-time, $\texttt{pay}_{ij(t-1)}$ indicates whether the previous episode was purchased, and X contains narrative features.

Heterogeneity. Each user i belongs to a latent segment $z \in \{1, \ldots, K\}$ with $K = 4$. Segment membership is shared across all series consumed by the same user.

Objective. Recover segment-specific reward functions $r^*(s, a, z)$ and policies $\pi^*(a \mid s, z)$ that rationalize observed trajectories under the maximum-entropy framework:

$$\max_{\pi}\ \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \big(r(s_t, a_t, z) + \mathcal{H}(\pi(\cdot \mid s_t, z))\big)\right], \quad \gamma = 0.99.$$

Termination. Exit ($a = 2$) or series completion ($g = 1$). Both lead to absorbing state s_{abs} with $r(s_{\text{abs}}, a, z) = 0$ and $V(s_{\text{abs}}) = 0$.

Networks.

Component	Parameterizes	Architecture
$g_\theta(s, a, z)$	Immediate reward	2-layer MLP (128, 64), Leaky ReLU
$h_\phi(s, z)$	Shaping potential $\approx V(s)$	2-layer MLP (128, 64), Leaky ReLU
$\pi_\omega(a \mid s, z)$	Policy (CCPs)	2-layer MLP (128, 64), Leaky ReLU
$q_\psi(z \mid \tau)$	Posterior (segment assignment)	Bi-LSTM (128) + MLP (256, 64)

Discriminator structure. The discriminator embeds a Bellman-consistent decomposition:

$$f_{\theta,\phi}(s, a, s', z) = g_\theta(s, a, z) + \gamma\, h_\phi(s', z) - h_\phi(s, z), \tag{1}$$

$$D_{\theta,\phi}(s, a, s', z) = \frac{\exp\big(f_{\theta,\phi}(s, a, s', z)\big)}{\exp\big(f_{\theta,\phi}(s, a, s', z)\big) + \pi_\omega(a \mid s, z)}. \tag{2}$$

Note: ϕ parameterizes the shaping potential h, not the Q-function. The policy π_ω is a separate network. The Q-function is never directly parameterized.

Symbol	Meaning
τ	Trajectory $(s_0, a_0, s_1, a_1, \ldots)$ for one user in one series
$\mathcal{D}_{\text{obs}}$	Set of observed transitions (s, a, s') from platform data
$\mathcal{D}_{\text{sim}}$	Set of simulated transitions from current policy π_ω
f	Discriminator logit (real-valued score)
D	Discriminator output $\in [0, 1]$ (probability transition is observed)
α	Learning rate for gradient descent
ε	Convergence tolerance for stopping criterion
λ_{CR}	Weight on within-user consistency regularization (S3)
λ_{MI}	Weight on mutual information reward (S4); set to 0 after posterior converges

Additional notation. We suppress subscripts ijt in the algorithm for clarity.

What each step does. **S1** generates synthetic reading behavior by running the current policy forward through actual book content. **S2** is the core step: it updates the reward and value networks so the discriminator can tell real reading behavior from fake. This is where the structural parameters θ (reward) and ϕ (value) are learned. **S3** assigns each user to a latent segment, enforcing that the same user behaves consistently across different series. **S4** converts the discriminator's output into a scalar reward signal for each simulated transition. **S5** updates the policy to maximize these rewards using PPO. This is the forward RL step that would be soft value iteration in a tabular MDP but requires PPO because the state space is high-dimensional.

Algorithm: LSW AIRL with Heterogeneity

Initialize parameters $\theta_0, \phi_0, \omega_0, \psi_0$

S1. Simulate Trajectories *(generates $\mathcal{D}_{sim}$)*

for $z = 1, \ldots, K$ **do**
  Sample series j, set initial state s_0
  Roll out $\pi_\omega(a \mid s, z)$ episode-by-episode to generate τ^{sim}
  Stop on exit $(a{=}2)$ or series completion $(g{=}1) \rightarrow$ enter s_{abs}
end for

S2. Discriminator Update *(updates θ, ϕ)*

for each (s, a, s') in $\mathcal{D}_{\text{obs}} \cup \mathcal{D}_{\text{sim}}$ **do**
  $f \leftarrow g_\theta(s, a, z) + \gamma\, h_\phi(s', z) - h_\phi(s, z)$
  $D \leftarrow \exp(f) \,/\, \big(\exp(f) + \pi_\omega(a \mid s, z)\big)$
end for
$\mathcal{L}_{\text{disc}} \leftarrow -\frac{1}{|\mathcal{D}_{\text{obs}}|} \sum_{\mathcal{D}_{\text{obs}}} \log D \;-\; \frac{1}{|\mathcal{D}_{\text{sim}}|} \sum_{\mathcal{D}_{\text{sim}}} \log(1 - D)$
$\theta, \phi \leftarrow \theta, \phi - \alpha\, \nabla_{\theta,\phi}\, \mathcal{L}_{\text{disc}}$

S3. Posterior Update *(updates ψ)*

for each trajectory τ_i **do**
  Compute $q_\psi(z \mid \tau_i)$
end for
$\mathcal{L}_{\text{post}} \leftarrow \mathcal{L}_{\text{classify}} + \lambda_{\text{CR}} \cdot \mathcal{L}_{\text{consistency}}$ (same user across series $\rightarrow$ similar z)
$\psi \leftarrow \psi - \alpha\, \nabla_\psi\, \mathcal{L}_{\text{post}}$

S4. Reward Assignment *(no parameter updates)*

for each simulated (s, a, s') with segment z **do**
  $\tilde{r}_{\text{adv}} \leftarrow f_{\theta,\phi}(s, a, s', z) - \log \pi_\omega(a \mid s, z)$
  $\tilde{r}_{\text{MI}} \leftarrow \log q_\psi(z \mid \tau)$ (standardized)
  $\tilde{r} \leftarrow \tilde{r}_{\text{adv}} + \lambda_{\text{MI}} \cdot \tilde{r}_{\text{MI}}$
end for
After posterior converges: freeze q_ψ, set $\lambda_{\text{MI}} = 0$.

S5. Policy Update via PPO *(updates ω)*

for $z = 1, \ldots, K$ **do**
  $\omega \leftarrow \text{PPO}(\omega;\ \max_{\pi_\omega}\ \mathbb{E}[\sum_t \gamma^t\, \tilde{r}])$ (clipped surrogate objective)
end for

Check: $\max_{s,a,z} |\pi_\omega^{\text{new}}(a \mid s, z) - \pi_\omega^{\text{old}}(a \mid s, z)| < \varepsilon$? Stop. Else repeat from S1.

repeat

At Convergence

When $D = 1/2$ everywhere (discriminator cannot distinguish observed from simulated):

$$f_{\theta,\phi}(s, a, s', z) = \log \pi^*(a \mid s, z) = A^*(s, a, z)$$

where A^* is the advantage function. With the anchors $g_\theta(s, \text{exit}, z) = 0$ for all s, z and $h_\phi(s_{\text{abs}}, z) = 0$, the decomposition is unique:

$$g_\theta(s, a, z) = r^*(s, a, z), \qquad h_\phi(s, z) = V^*(s, z).$$